

Vedant Desai

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EDUCATION

University of Michigan

B.S. in Computer Science and Economics

Expected May 2028

Ann Arbor, MI

- **GPA:** 3.7 / 4.0
- **Coursework:** Data Structures & Algorithms, Linear Algebra for Machine Learning, Microeconomics, Macroeconomics, Discrete Mathematics, Calculus III, Physics (Mechanics)

EXPERIENCE

Lyndbrook Capital

Data Engineering Consultant

Feb 2026 – Apr 2026

Remote

- Contracted pre-LOI to build acquisition intelligence for a boutique search fund targeting water utility operators; aggregated EPA ECHO and MassGIS regulatory data into a unified PWSID entity database and delivered 800+ validated Day-1 acquisition targets within the engagement window.
- Built a Review Velocity scoring algorithm that proxied fleet expansion and operational scale from public compliance data, filtering 800 acquisition targets down to a 280-lead qualified shortlist — a 35% precision rate against the fund's revenue criteria.

Vylet

Founder — Automated lead-sourcing platform for PE/search-fund; live product generating \$1,500 MRR across three clients

May 2026 – Present | vyletdata.com

- Engineered a LangSmith eval pipeline spanning 20 adversarial business test cases across 13 archetype labels — manufacturers, SaaS tools, PE holding companies, geographic mismatches — then layered deterministic Pydantic consensus gates to lift extraction faithfulness from 50% to 90%.
- Launched Vylet, automating a ~30-minute manual process per business into a Dockerized LangGraph pipeline generating 30 scored leads in 30 minutes — a 30x speedup — with Redis/Celery workers on a recurring cycle.

Michigan Data Consulting (MDC)

Data Engineer — Michigan Campaign Finance Network

Jan 2026 – May 2026

Ann Arbor, MI

- Replaced manual Michigan campaign-finance research — portal searches capped by the Bureau of Elections, irregular Excel exports, hand normalization at ~2 hours per committee — with a Requests + Pandas ETL that ingests filings directly, eliminating ~800 hours of manual pulls across 400 tracked PACs.

PROJECTS

Granular Synthesizer Plugin | C++, JUCE

Real-time audio DSP plugin built from scratch in C++/JUCE — sine oscillator through full granular engine with lock-free threading and release-ready VST3/AU binaries

github.com/Verdent06/granular-synth

- Enforced the audio thread's zero-allocation constraint by pre-allocating a `MemoryPool<Grain, 64>` slab per voice at plugin startup — a fixed free-list that acquires and releases grain slots without touching the heap — so `processBlock()` never calls `new` or `delete` after `prepareToPlay()` completes.
- Wired UI-to-audio delivery through a hand-rolled SPSC FIFO (64-slot fixed array, atomic head/tail with acquire/release ordering) so slider changes reach the audio thread without a mutex — and routed WAV handoff via atomic `shared_ptr` swap so the UI thread never blocks `processBlock()`.

SignalWeaver | Python, LoRA, Docker, PostgreSQL, GitHub Actions

Multi-signal financial research platform combining fundamentals, macro indicators, and news sentiment (research assistant; not investment advice)

github.com/Verdent06/SignalWeaver

- Lifted financial-sentiment classification accuracy from 81% to 96% by LoRA fine-tuning a quantized meta-llama/Meta-Llama-3.1-8B-Instruct model on 3,454 Financial PhraseBank entries, evaluated on a held-out test set.
- Containerized the stack with Docker Compose (API + Postgres/pgvector + nginx frontend) and a GitHub Actions CI pipeline (frontend build, pytest, API image build on main).

TECHNICAL SKILLS

Languages Python, C++, SQL

AI / ML LangSmith, LangGraph, Pandas

Cloud & Infrastructure Docker, Git, GitHub Actions